

Expansion Planning of Soft Open Points Based Distribution System Considering EV Traffic Flow

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Abstract—Regarding distribution systems coupled with transportation networks, this study proposes an expansion planning approach to minimize various costs with respect to soft open point (SOP) based distribution system and electric vehicle (EV) navigation. An energy consumption model of EV that considers traffic flow is developed to analyze charging behaviors of EVs. Subsequently, the EV average speed and related energy consumption is estimated to generate optimal charging routes. The proposed model of the EV energy consumption can help to minimize the cost of navigating EVs along the transportation network according to the traffic flow. Referring to the distribution system, the total cost associated with the investment and maintenance of network assets as well as subsequent operations is optimized. Hereafter, an expansion planning model of distribution system considering EV navigation is established. A relaxation approach is employed through which the prescribed nonconvex mixed-integer nonlinear planning model is converted into the form of mixed-integer second order cone programming that can be efficiently solved by commonly used solvers. A 54-bus distribution system coupled with a 35-node transportation network is employed in the case study section to verify the effectiveness of the proposed method. The rationality of considering SOP deployment and EV navigation is validated.

Index Terms—Distribution system planning, EV navigation, soft open points, traffic flow, transportation network.

NOMENCLATURE

A. Set

$\Psi_l, \Psi_{sc}, \Psi_{se}$ Candidate sets of lines, substation construction and substation extension.

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$\Psi_{sop}, \Psi_{ess}, \Psi_{pv},$

Ψ_{cs}

B. Parameters

$TEV_{x,y}^0$

$C_{x,y}$

A_i^{SOP}

$T, \bar{h}, N$

σ_i^{pv}

$P_{i,t}^{DSM}$

τ_t

r

$S_{i,rated}^{ext}$

N_{root}, ε

C^{sop}, C^{pv}

$C^{ESS,P}, C^{ESS,E}$

$C^{CS,P}, C^{CS,E}$

$S_{i,rated}^{Sub}, S_{i,rated}^{OSub}$

μ_{cha}, μ_{dis}

$A_d, C_r, C_d, \lambda(t)$

V_{ij}, C_{ij}, δ

$C_{ij}^L, C_i^{SC}, C_i^{se}$

C. Variables

$V_{(x,y)}$

F_{res}

Candidate sets of SOP construction, ESS construction and PV construction. Candidate set of CS construction.

Travel time of edge (x, y) at the maximum speed.

Road capacity of Edge (x, y).

Loss coefficient of SOP at node i.

Observation time, average headway of between vehicles and total observed quantity of vehicles.

Maximum ratio of PV's power curtailment.

Permitted load curtailment of node i at time t.

Time period of each scenario.

Discount rate.

Available capacity of substation extension at node i.

Number of root nodes, a variable that would be injected with the power of all the root nodes.

Investment cost for SOP and PV per unit capacity.

ESS investment cost per unit capacity and per unit power.

CS investment cost per unit capacity and per unit power.

Capacity of candidate substations and substations that already exist.

Charging and discharging efficiency of ESS.

Frontal area of the vehicle, rolling coefficient, drag/air coefficient and road slope.

Traffic volume, capacity of edge ij and road saturation.

Line construction cost per unit length, substation construction and extension cost.

Average EV speed regarding traffic flow of edge (x, y).

Driving resistance of the vehicle.

$TOR_{(x,y)}$	Average time occupancy rate of edge (x, y).
$D_{x,y}$	The traffic density (number/h).
D_u	Probability of each scenario.
$\Phi_{mn}^{SOP,L}$	Status of branch $m-n$.
$P_{i,t}^{SOP,L}$	Loss of SOP active power at node i .
θ^L	Power angle for nodes with load.
X_{ij}^L	The decision variables for line construction.
$E_{(x,y),t}, EC$	Estimated energy consumption, energy consumption of all related routes.
X_i^{SC}, X_i^{SE}	The decision variables for substation construction and extension.
S_i^{sop}, S^{CS}	The decision variables for quantity of SOP/CS installation.
E^{ESS}, E^{CS}	Installation power and capacity for the ESS and CS.
$P_{ij,t}, Q_{ij,t}$	Active/reactive power flow of branch ij .
$P_{i,t}^{inj}, Q_{i,t}^{inj}$	Injected active and reactive power at node i .
$P_{i,t}^{Sub}, P_{i,t}^{SOP}, P_{i,t}^{PVG}$	Active power generated by substations, SOP and PV.
$P_{i,t}^{ESS}, P_{i,t}^{CS}$	Active power generated by ESS, CS.
$Q_{i,t}^{Sub}, Q_{i,t}^{SOP}, Q_{i,t}^{PVG}$	Reactive power generated by substations, SOP and PV.
$Q_{i,t}^{ESS}, Q_{i,t}^{CS}$	Reactive power generated by ESS, CS.
$P_{i,t}^{ESS,cha}, P_{i,t}^{ESS,dis}$	Charging and discharging power of ESS located at node i .

I. INTRODUCTION

WITH climate change and carbon emission concerns, more awareness is turning to the reduction of fossil fuel consumptions. Numerous efforts of various strategies aimed at addressing these concerns have begun to show positive effects. Performed as an important role of realizing environmental topics, the development of electric vehicle (EV) related technologies has been promoted by various policy supports in many countries [1], [2]. Affected by traffic flow, EV charging stations are deeply coupled in distribution systems implies that the EV performance could also be related to the functionality of distribution systems. The EV performances could not only determine the planning of charging stations but also benefit the mitigation of various planning costs [3], [4]. Hereafter, considering EV related distribution systems, the planning approaches that address features of both distribution system and transportation network should be extensively studied.

To address the issue of EV related distribution systems, various planning approaches have been developed. A multiple-charger multiple-port planning method is developed to promote the capability of scheduling limited chargers to serve more EV [5]. To reduce investment/ operation cost, literature [1], [6] models the planning of distribution systems with charging stations as a multi-objective multi-stage problem, while the charging demand at all stations is maximized. In [7], planning strategies are presented to address network planning with renewable energy and service of charging stations, which can not

only guarantee the charging service but also minimize planning cost. Aiming to minimize the weighted sum of various costs, energy loss and waiting time of shared EVs, a planning model for distributed system with renewable generations is proposed in [8]. However, studies above that investigate the planning of distribution systems with EV charging stations ignore the nexus of distribution system and transportation network.

Subsequently, regarding the nexus of distribution system and transportation network, the increasing prevalence of accessing EV in system planning raise the awareness of estimating EV energy consumptions. Research [9] presents a planning approach for EV dynamic wireless charging system, in which the economic effects of wireless charging is maximized. And a Nesterov's model with multiple traffic patterns is deployed in the transportation network to solve the traffic flow related assignment problem. Research [10] studies the smart scheduling for EVs with limited charging infrastructures. The proposed approach minimizes the total travel time for single EV in traffic flow. To mitigate negative impacts on traffic conditions caused by applied charging stations, research [11] investigates the planning strategy for allocating EVs with balanced traffic flow. And in summary, regarding planning of distribution systems coupled with transportation network, the inadequacy of the extant approach is, the impact of traffic flow and subsequent EV navigation is in lack of quantification. Thus, the distribution system considering EV navigation should be planned and operated in a more coordinated fashion.

Hereafter, considering EV navigation, a rational solution to mitigate various costs of distribution system planning can be the deployment of SOP. Enabling the transfer of active power between networks and compensate reactive power, the soft open point (SOP) is consisted of voltage source converters connected by capacitors. The deployment of SOP can offer active power flow management, reactive power compensation and voltage regulation [12], [13]. And comparing to conventional distribution systems, the SOP based distribution system would obtain lower loss, voltage profile enhancement, and system dependability improvement. Based on SOP, the voltage control can be applied to mitigate operation costs and voltage violations of renewable power contained distribution systems [14]. The dispatching of SOP coordinated with energy storage and EV management would enhance system flexibility and the maximum hosting capacity of photovoltaic (PV) generation [13], [15].

On the premises of the analysis above, regarding distribution systems coupled with transportation networks, an expansion planning approach that addresses the impact of traffic flow and subsequent EV navigation should be developed. Herein, based on the deployment of SOP, this study proposes an expansion planning approach for distribution systems that considers the EV navigation affected by traffic flow. The main contributions of this study are summarized as follows:

- Referring to dynamic traffic flow, a combined assessment model that extracts features of both traffic conditions and distribution systems is proposed. Subsequently, the EV navigation cost could be more precisely addressed.

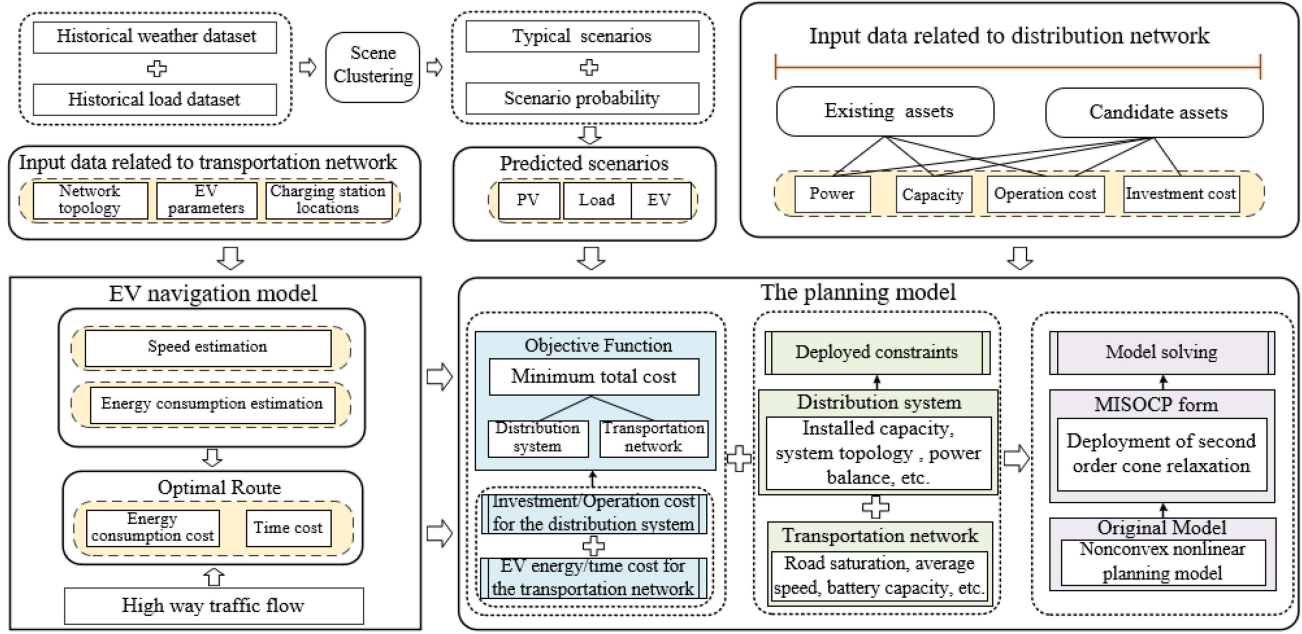


Fig. 1. The schematic of the proposed model.

- On the basis of extra flexibility provided by prescribed SOP employment, the planning cost and energy consumption of EV navigation can be further mitigated, while the capability of renewable energy accommodation would be enhanced.

The overall schematic of this study is illustrated in Fig. 1. The rest of this study is organized as: Section II introduces the EV navigation model referring to traffic flow, Section III presents the detailed planning model, Section IV refers to the result analysis, and Section V concludes the whole study.

II. EV NAVIGATION REGARDING TRAFFIC FLOW

Regarding the coupling of transportation networks and distribution systems, herein, to characterize EV navigation cost, this section classifies energy consumption for each route. The transportation network can be formulated as a graph G that constructed by vertices and edges (V, G) . Subsequently, regarding deployed traffic flow of graph G , this section provides an EV navigation model that investigates 1) EV navigation's average speed, 2) EV navigation cost per edge, 3) Optimal EV navigation route/edges.

A. Average Speed Estimation

Since the energy consumption of EV is primarily determined by vehicle speed, special consideration should be turned to speed estimation along the EV navigation route. And although the obtained information of traffic flow (i.e., traffic volume travel time and traffic density) could indicate the mean vehicle flow speed alongside an edge, the individual EV drivers may show characteristic deviations from this flow speed. Thus, an individual average speed estimation model is provided.

Let $EEV_{x,y}$ denote the number of arriving EVs on the edge (x, y) within the time interval T , based on the unit of hour and

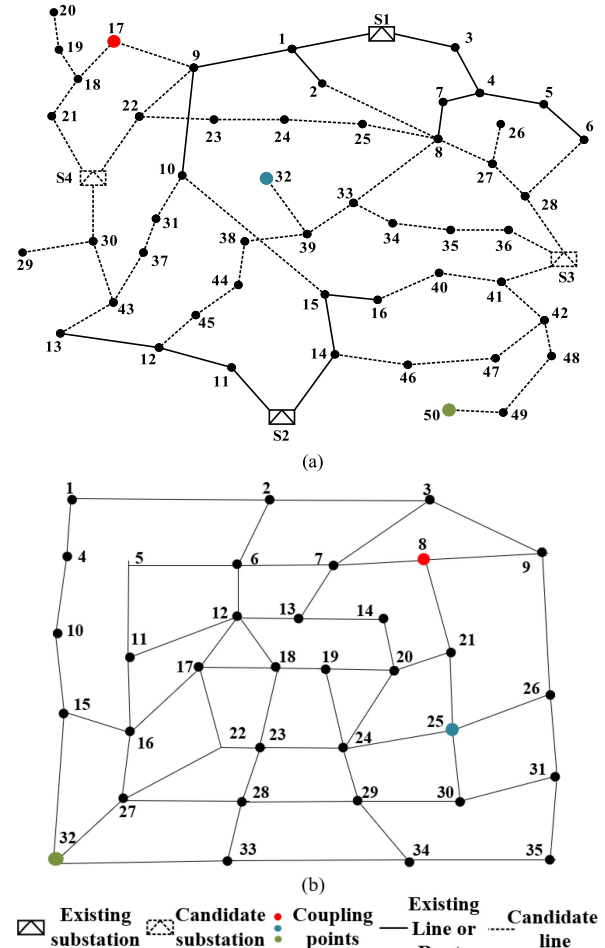


Fig. 2. Test systems. (a) Deployed distribution system. (b) Deployed transportation system.

traffic volume, the travel time for vehicles on the edge (x, y) can be written as [16],

$$TEV_{x,y} = TEV_{x,y}^0 + 0.15TEV_{x,y}^0 (EEV_{x,y}/C_{x,y})^4 \quad (1)$$

where $TEV_{x,y}^0$ is the travel time of edge (x, y) at the maximum speed regarding no congestion conditions; $C_{x,y}$ is the road capacity of the edge.

Referring to the EV speed estimation, average time occupancy rate is a common indicator of traffic flow that can be derived from the density of an edge, one of the information that directly obtained from traffic flow. And since the accumulation of vehicles boosts the congestion level of road segments, and the congestion level of a road is directly related to the vehicle speed, so the estimation of EV average speed must consider the vehicle length. The mathematical relationship between these two indicators is as follow [2],

$$T = N\bar{h} \quad (2)$$

$$TOR_{(x,y)} = \frac{1}{T} \sum_n t_n = ((1/N) \sum_n l_n/v_n)/\bar{h} + dD_{x,y} \quad (3)$$

where $TOR_{(x,y)}$ is the average time occupancy rate of edge (x, y) that can be derived based on the traffic density $D_{x,y}$ (number/h).

If employing an assumption that the length of EV is fixed at l , the $TOR_{(x,y)}$ can be derived as

$$TOR_{(x,y)} = (l + d)D_{x,y} \quad (4)$$

And since $EEV_{(x,y)} = N/T = 1/\bar{h}$, the average EV speed $V_{(x,y)}$, representing the average EV speed regarding traffic flow of edge (x, y) , can be expressed as

$$V_{(x,y)} = EEV_{(x,y)} \times (l + d)/TOR_{(x,y)} \quad (5)$$

Consequently, the average EV speed can be seen as only determined by $TOR_{(x,y)}$.

In general, the estimation of $V_{(x,y)}$ can be summarized as: a) Derive travel time of edge (x, y) $EEV_{x,y}$ from $C_{x,y}$; b) Derive average time occupancy rate $TOR_{(x,y)}$ from $D_{x,y}$; c) Obtain average EV speed $V_{(x,y)}$ based on road saturation.

B. Energy Consumption Estimation

On the premises of the speed estimation method above, an estimation model for EV energy consumption can be deployed following the proposed speed estimation model.

The general longitudinal equation of vehicle motion can be written as

$$F(t) = m \frac{dv(t)}{dt} + F_{res}(t) \quad (6)$$

where the F refers to the required force of vehicle movement; F_{res} refers to the driving resistance.

The further expression of F_{res} can be denoted as [17], [18]

$$\begin{aligned} F_{res}(t) &= F_{roll}(t) + F_{dra}(t) + F_{gra}(t) \\ F_{roll}(t) &= mgC_r \cos(\lambda(t)) \\ F_{dra}(t) &= 1/2 A_d C_d \alpha v^2(t) \\ F_{gra}(t) &= mg \sin(\lambda(t)) \\ E_{(x,y),t} &= 1/3600 F_{res}(t) V_{(x,y),t} \Delta t \end{aligned} \quad (7)$$

Owing to the fact that the battery internal loss only obtains 3% to 6% of the overall energy consumption, this study only considers the energy consumption of express way routing [17], the final energy consumption for an edge and all related routes can be derived as

$$EC = \sum_i E_i(t_i) \quad (8)$$

where $E_i(T)$ is the energy consumption for an edge while EC represents the energy consumption of all related routes.

The deviation of EC can be summarized as: a) Obtain the general driving resistance F_{res} for each edge; b) Derive the estimated energy consumption $E_{(x,y),t}$ during time period Δt ; c) Obtain energy consumption for each edge $E_i(T)$ and subsequent all related routes EC .

C. EV Charging Optimal Route

Regarding the prescribed EV speed estimation and subsequent energy consumption models, the EV optimal route regarding traffic flow can be established then. For the optimal route searching, A* algorithm is a commonly applied algorithm to solve this problem [10], [19]. The reasons that A* algorithm is appropriate to be deployed here are: A* is a commonly applied algorithm with efficient implementation for a variety of systems that would guide the deployment of proposed solutions. In addition, the formulated minimum energy consumption/time searching is a linear problem and can be decomposed from the following planning model. Lastly, the deployment of this heuristic function is convenient.

Conventional A* algorithms only choose a path refers to the lowest weight without accounting for constraints. However, the relating paths that violating prescribed constraints actually can be avoided by deploying further testing when exploring neighboring nodes [19]. Hence, the model of EV optimal route not only considers the lowest cost but also subject to transportation network's constraints. And it is ensured that the proposed algorithm would generate an effective solution that the found path is with the lowest cost. The algorithm's result returns an estimation of both time cost and the energy consumption between start node and destination node. The edge is only feasible if the charging station selection and acceptable saturation constraints are satisfied.

In summary, the constraints followed by EV optimal route are threefold. Firstly, each EV only chooses a charging station based on which the corresponding route is selected. Let T_c denotes the decision variable of charging station in the transportation network, set R_c as the decision variable of charging route. The charging station selection constraints would be

$$\sum T_c = 1; \sum R_c = 1 \quad (9)$$

Then, transportation and distribution network coupled constraints, can be written as

$$\sum T_c D_c = 1; \sum D_c = 1 \quad (10)$$

At the last stage, considers the traffic volume V_{ij} , capacity C_{ij} of edge ij and driver acceptable saturation R , the road saturation δ , which characterize the traffic condition of a single

Algorithm 1: Pseudocode for A* Search.

Closed list = []
 Open list = [Start Node n_0]
Input:
 C the energy consumption from start node to others;
 P the parents or predecessors of the prescribed node;
 $C(n_0) = 0$;
for $i = 1$ to $n - 1$
 if current node not in open and closed list
 add current node to open list and calculate heuristic set
 as its parents/predecessors;
end
for edge $(x, y) \in R(G)$
 if $C(x) > C(y) + C(x + y)$;
 $C(x) = C(y) + C(x + y)$;
 $P(x) = y$;
end
end for
return optimal route
end for

road would be

$$\delta = \frac{V_{ij}}{C_{ij}} \times 100\% \quad (11)$$

$$\sum Pr \{V_{ij} \leq R_c \times \delta C_{ij}\} \geq R \quad (12)$$

The pseudocode for A* searching is shown in Algorithm 1, in which $R(G)$ represents for the edges (routes) and nodes of the transportation network.

Herein, the information obtained from traffic flow contains traffic density $D_{x,y}$, observation time T and quantity of vehicles N . The prescribed information would then be applied to estimate the average vehicle speed. After that, T, N would be engaged to decide the traffic volume V_{ij} , by which the road saturation δ can be derived.

According to the employed traffic flow model, it should be noted that the arriving vehicles would affect the road saturation, while the road saturation would determine the vehicle speed, which in turn would affect the arriving vehicles. Hence, by ensuring vehicle speed and each road's saturation, the traffic flow model can then denote the optimal charging route of navigated EVs with minimum energy and time cost. Consequently, an optimal route that obtains minimum energy and time cost on the transportation network is established. And to summarize, the EV navigation model of Section II would estimate EV charging behaviors regarding dynamic traffic flow. Hereafter, the number of EV at each charging point with related charging demand can be provided to the subsequent planning model.

III. SOP BASED DISTRIBUTION SYSTEM PLANNING MODEL

In this section, the proposed method develops an integrated planning model of distribution systems. Regarding the charging information provided by the model of Section II, this model determines what expansion actions should be taken to minimize

total cost. The objective function and related operation constraints of the proposed model is introduced. And the second order cone relaxation of the proposed planning optimization problem is analyzed.

A. Objective Function

The planning model takes the minimum annual cost as the objective function. In addition, CS and ESS in this section are related to charging stations and energy storages of the system. Referring to investment cost C_{inv} and operation cost C_{ope} , the planning model can be denoted as

$$\min f = C_{inv} + C_{ope} \quad (13)$$

$$\rho = \frac{r(1+r)^i}{(1+r)^i - 1} \quad (14)$$

$$\begin{aligned} C_{inv} = & \rho_L \sum_{ij \in \Psi_L} C_{ij}^L X_{ij}^L + \rho_{sc} \sum_{i \in \Psi_{sc}} C_i^{SC} X_i^{SC} \\ & + \rho_{se} \sum_{i \in \Psi_{se}} C_i^{se} X_i^{se} + \rho_{sop} \sum_{i \in \Psi_{sop}} C_i^{sop} S_i^{sop} \\ & + \rho_{ESS} \sum_{i \in \Psi_{ess}} (C^{ESS,P} S^{ESS} + C^{ESS,E} E^{ESS}) \\ & + \rho_{CS} \sum_{i \in \Psi_{cs}} C^{CS,E} E^{CS} \\ & + \rho_{pv} \sum_{i \in \Psi_{pv}} C^{pv} S_i^{pv} \end{aligned} \quad (15)$$

$$\begin{aligned} C_{ope} = & \sum_{u \in \Omega_u} D_u \sum_{t \in \Omega_t} \tau_t \left(\sum_{i \in \Psi_{se} \cup \Psi_{sc}} C_{i,u,t}^{Buy} P_{i,u,t}^S \right. \\ & \left. + \sum_{i \in \Psi_{pv}} C_{i,u,t}^{pv,c} P_{i,u,t}^{pv,c} \right) \end{aligned} \quad (16)$$

where objective function (13) minimizes both investment cost (15) and operation cost (16); the D_u and τ_t are the probability and time period of each scenario; r is the discount rate that determines the capital recovery factor ρ .

B. Construction Constraints

Regarding distribution systems, the network assets would follow the construction constraints below

1) Topological Constraints:

$$\begin{aligned} \sum_{mn \in B} \Phi_{mn} &= N_{node} - N_{root}, \Phi_{mn} \in \{0, 1\} \\ \begin{cases} P_{j,t}^* = \sum_k P_{jk,t}^* - \sum_k (P_{ij,t}^* - R_{ij,t} \ell_{ij}^*) \\ \quad + g_j V_{j,t}^* = \varepsilon \\ Q_{j,t}^* = \sum_k Q_{jk,t}^* - \sum_k (Q_{ij,t}^* - X_{ij,t} \ell_{ij}^*) \\ \quad + b_j V_{j,t}^* = \varepsilon \end{cases} \end{aligned} \quad (17)$$

where $P_{j,t}^*$ and $Q_{j,t}^*$ confirm that the nodes with injected power would have at least one connection branch with its injector.

2) PV Constraints:

$$\begin{cases} \sum_{i \in \Psi_{pv}} U_i^{pv} P_{unit}^{pv} \leq \varphi \sum_{i \in \Psi_N} P_{i,Max}^{Load} \\ 0 \leq U_i^{pv} \leq U_{i,Max}^{pv} \end{cases} \quad (18)$$

where P_{unit}^{pv} is the installation capacity of PV per unit; φ determines the allowed penetration of the distribution systems; $U_{i,Max}^{pv}$ is the maximum number of permitted PV installation at node i .

3) ESS Constraints:

$$\begin{cases} \sum_{i \in \Psi_{ESS}} E_i^{ESS} \leq E_{i,Max}^{ESS} \\ \sum_{i \in \Psi_{ESS}} S_i^{ESS} \leq S_{i,Max}^{ESS} \\ 0 \leq E_i^{ESS} \leq E_{i,Max}^{ESS} \\ 0 \leq S_i^{ESS} \leq S_{i,Max}^{ESS} \end{cases} \quad (19)$$

where E_i^{ESS} , S_i^{ESS} represent ESS's capacity and power installation located at node i .

4) CS Constraints:

$$\begin{cases} \varepsilon P_i^{CS} \leq E_i^{CS} \leq E_{i,Max}^{CS} \\ \sum_{i \in \Psi_{CS}} E_i^{CS} \leq E_{i,Max}^{CS} \end{cases} \quad (20)$$

where E_i^{CS} represents CS's capacity installation located at node i ; ε represents the number of charging vehicles at node i .

5) SOP Constraints:

$$\begin{cases} 0 \leq U_{ij}^{SOP} \leq U_{ij,max}^{SOP} \\ S_{ij}^{SOP} = U_{ij}^{SOP} S_{ij}^{SOP} \end{cases} \quad (21)$$

In (21), $U_{i,t}^{SOP}$, $U_{j,t}^{SOP}$ denotes the quantity of installed SOP; $S_{ij,t}^{SOP}$ refer to SOP's capacity.

C. Operational Constraints

In this section, system operational constraints of the deployed assets are introduced.

1) Power Balance Constraints:

$$V_{i,t} l_{ij,t} - P_{ji,t}^2 - Q_{ji,t}^2 = 0 \quad (22)$$

$$\sum_j P_{ij,t} - \sum_k (P_{i,t} - R_{ki} l_{ki,t}) = I P_{i,t}$$

$$\sum_j Q_{ij,t} - \sum_k (Q_{i,t} - X_{ki} l_{ki,t}) = I Q_{i,t}$$

$$\begin{aligned} P_{i,t}^{inj} &= P_{i,t}^{Sub} + P_{i,t}^{SOP} + P_{i,t}^{PV} + P_{i,t}^{ESS} \\ &\quad + P_{i,t}^{CS} - P_{i,t}^L + P_{i,t}^{DSM} \\ Q_{i,t}^{inj} &= Q_{i,t}^{Sub} + Q_{i,t}^{SOP} + Q_{i,t}^{PV} - Q_{i,t}^L + Q_{i,t}^{DSM} \end{aligned} \quad (23)$$

where $P_{ij,t}$, $Q_{ij,t}$ is the active/reactive power flow of branch ij ; $P_{i,t}^{inj}$ and $Q_{i,t}^{inj}$ describes the injected active and reactive power at node i .

2) Substation Constraints:

$$\begin{cases} \sqrt{(P_{i,t}^{Sub})^2 + (Q_{i,t}^{Sub})^2} \leq x_i^{Sub} S_{i,rated}^{Sub} \\ \sqrt{(P_{i,t}^{Sub})^2 + (Q_{i,t}^{Sub})^2} \leq S_{i,rated}^{OSub} + x_i^{ext} S_{i,rated}^{ext} \end{cases} \quad (24)$$

where $S_{i,rated}^{Sub}$, $S_{i,rated}^{OSub}$ refers to the capacity of candidate substations and substations that already exist respectively; $S_{i,rated}^{ext}$ describes the available capacity of substation extension at node i .

3) Node Voltage and Branch Current Constraints:

$$\begin{cases} V_{i,Min}^2 \leq V_{i,t}^2 \leq V_{i,Max}^2 \\ X_{ij} I_{max,ij}^2 \geq I_{ij,t}^2 \end{cases} \quad (25)$$

where $V_{i,t}^2$ represents for the square value of voltage at node i with upper and lower bounds; $I_{max,ij}^2$ refers to the square value of current of branch ij .

4) PV Constraints:

$$\begin{cases} P_{i,Min}^{pv} \leq P_{i,t}^{pv} \leq P_{i,Max}^{pv} \\ \sqrt{(P_{i,t}^{pv})^2 + (Q_{i,t}^{pv})^2} \leq z_i^{pv} P_{i,t}^{rated} \\ 0 \leq P_{i,t}^{pv} \leq \sigma_i^{pv} P_{i,t,Max}^{pv} \end{cases} \quad (26)$$

where $P_{i,Min}^{pv}$ and $P_{i,Max}^{pv}$ refer to lower and upper boundary of PV's active power; σ_i^{pv} determines the maximum ratio of PV's power curtailment.

5) SOP Constraints:

$$P_{i,t}^{SOP} + P_{j,t}^{SOP} + P_{i,t}^{SOP,L} + P_{j,t}^{SOP,L} = 0 \quad (27)$$

where $P_{i,t}^{SOP,L}$ denotes the loss of SOP active power at node i ; $A_{i,t}^{SOP}$ denotes the loss coefficient of SOP at node i .

$$\begin{cases} A_{i,t}^{SOP} \sqrt{(P_{i,t}^{SOP})^2 + (Q_{i,t}^{SOP})^2} = P_{i,t}^{SOP,L} \\ A_{j,t}^{SOP} \sqrt{(P_{j,t}^{SOP})^2 + (Q_{j,t}^{SOP})^2} = P_{j,t}^{SOP,L} \end{cases} \quad (28)$$

$$\begin{cases} \sqrt{(P_{i,t}^{SOP})^2 + (Q_{i,t}^{SOP})^2} \leq S_{i,t}^{SOP} \\ \sqrt{(P_{j,t}^{SOP})^2 + (Q_{j,t}^{SOP})^2} \leq S_{j,t}^{SOP} \end{cases} \quad (29)$$

6) ESS Constraints:

$$\begin{cases} 0 \leq P_{i,t}^{ESS,dis} \leq P_{i,t}^{ESS,Max} \\ 0 \leq P_{i,t}^{ESS,cha} \leq P_{i,t}^{ESS,Max} \\ E_{i,t+1}^{ESS} = E_{i,t}^{ESS} - \mu_{cha} P_{i,t}^{ESS,cha} - P_{i,t}^{ESS,dis} / \mu_{dis} \\ E_{i,t=1}^{ESS} = E_{i,t=24}^{ESS} \\ E_{i,Min}^{ESS} \leq E_{i,t}^{ESS} \leq E_{i,Max}^{ESS} \end{cases} \quad (30)$$

where $P_{i,t}^{ESS,cha}$ and $P_{i,t}^{ESS,dis}$ describe the charging and discharging power of ESS located at node i ; $E_{i,t}^{ESS}$ determines ESS's quantity of charge at time t .

7) Demand Side Management Constraints:

$$\begin{aligned} 0 &\leq P_{i,t}^{DSM} \leq P_{i,t,Max}^{DSM} \\ Q_{i,t}^{DSM} &= P_{i,t}^{DSM} \tan \theta^L \end{aligned} \quad (31)$$

where $P_{i,t}^{DSM}$ determines the permitted demand side management (DSM) of node i at time t ; θ^L is the power angle for nodes with load.

D. The MISOCP Transformation

Owing to the nonconvex nonlinear characteristics of the proposed planning model, herein, this study transforms it into a

mixed-integer second order cone programming form based on second order cone relaxations. Subsequently, the power balance constraints and (28) would be reformed as

$$\| [2P_{ij,t} 2Q_{ij,t} l_{ij,t} - v_{i,t}]^T \|_2 \leq l_{ij,t} + v_{i,t} \quad (32)$$

$$\begin{cases} \| [P_{i,t}^{SOP} Q_{i,t}^{SOP}]^T \|_2 \leq 2 \frac{P_{i,t}^{SOP}}{\sqrt{2A_{i,t}^{SOP}}} \frac{P_{i,t}^{SOP}}{\sqrt{2A_{i,t}^{SOP}}} \\ \| [P_{j,t}^{SOP} Q_{j,t}^{SOP}]^T \|_2 \leq 2 \frac{P_{j,t}^{SOP}}{\sqrt{2A_{j,t}^{SOP}}} \frac{P_{j,t}^{SOP}}{\sqrt{2A_{j,t}^{SOP}}} \end{cases} \quad (33)$$

Hereafter, denoted by (34), the proposed nonconvex nonlinear model is transformed into the form of second order cone programming that can be solved by commercial solvers.

$$\begin{aligned} \min f &= C_{inv} + C_{ope} \\ \text{s.t. } &(17) - (21), (23) - (27), (29) - (33) \end{aligned} \quad (34)$$

IV. CASE STUDY AND ANALYSIS

For the case and result analysis of this study, a 54-bus distribution system coupled with a 35-node (consisted of 57 routes) express way transportation network (8 km × 8 km) is deployed for effectiveness validation of the proposed planning approach. The proposed method in this study is implemented based on YALMIP optimization toolbox with MATLAB R2021a, and solved by GUROBI. The overall experimental process is computed on a PC with a processor of 12th Gen Intel (R) Core i5-12500 of 3.00 GHz and 16 GB RAM.

As for parameter settings of the transportation network, the EV charging stations are assumed to be positioned at nodes (8, 25, 32) of the transportation network. Subsequently, the EV battery capacity is assumed to be 60 kWh. The quantity of peak time navigated EVs that have charging demand is 200. And the initial SOC of navigated EV is randomly generated. The traffic flow related dataset is taken from [20]. The time cost included in the EV navigation cost is assumed to be a constant that is set as 5 \$/h. And herein, since the charging stations alongside express way transportation are normally located at positioned areas, this study does not consider the sitting of charging stations on the transportation network. The acceptable road saturation is assumed to be lower than 86%. Other navigation parameters for EV navigation are taken from [2], [21].

The total load of the distribution system is 70MW. The substations that can be extended are located at node 51, 52 while the substations that require constructions are located at node 53, 54 of the distribution system. The candidate nodes of EV charging stations are 17, 32, 50 of the distribution system. And the candidate nodes of ESS and PV are (8, 9, 13, 16) and (5, 8, 10, 26, 33, 37, 40) respectively. The candidate branches of SOP are (31-37, 33-34, 2-8, 10-15) of the distribution system. Considering the fluctuated feature of PV, on route EV and future load at the whole planning stage, herein, a cluster algorithm is applied in this study to deploy typical scenarios for the planning horizon.

Herein, the considered planning horizon is 20 years, and the discount rate is set as 6%. The candidate extension capacity of substation S1, S2 are 13.3 MW and 16.7 MW with corresponding cost of \$8 M and \$10 M. And the candidate constructed capacity of substation S3, S4 are 16.7 MW and 22.2 MW that corresponds

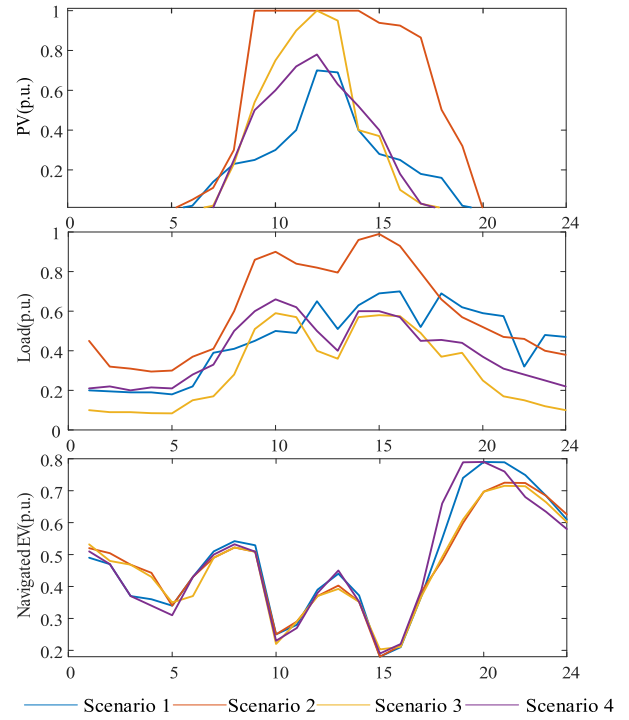


Fig. 3. Typical scenarios.

to the cost of \$14 M and \$20 M. The investment cost of distribution lines is 46×10^3 \$/km while the annual operation cost is 1×10^3 \$/km. Referring to SOP, the investment cost is 150 \$/kVA, the loss coefficient is 0.02. For PV, the rated capacity is set as 100 kW, with a unit capacity investment cost of 834 \$/kW. Furthermore, the unit operation cost for PV is 0.019 \$/kW while the unit curtailment cost for PV is 0.35 \$/kW, the penetration allowance of PV is 50%.

As shown in Fig. 3, four cluster-based typical scenarios of PV, load and navigated EV is applied for the computation of the presented model. It can be seen that the PV output is peaked at around 11 o'clock to 13 o'clock and disappeared at hours of night. Similarly, the general load of distribution system rises at 6 o'clock and declines after 15 o'clock. However, the largest peak of total navigated EV appears at night hours. It is also worth noticed that the deployed scenarios of navigated EV only refer to the number of on route EV that have charging demand, but not directly related to the charging demand.

The performance of the proposed approach is investigated through comparison of the results in the following case studies:

- Case A: Expansion planning that considers EV navigation and the deployment of SOP.
- Case B: Expansion planning that considers EV navigation with no SOP deployment.
- Case C: General expansion planning without consideration of SOP and EV navigation.

A. Expansion Planning Refers to EV Navigation and SOP

In case A, both EV navigation and SOP deployment are considered in the planning approach. Since the peak time navigated EV that has charging demand is set as 200, the number of EV at other time can be obtained according to typical scenarios shown

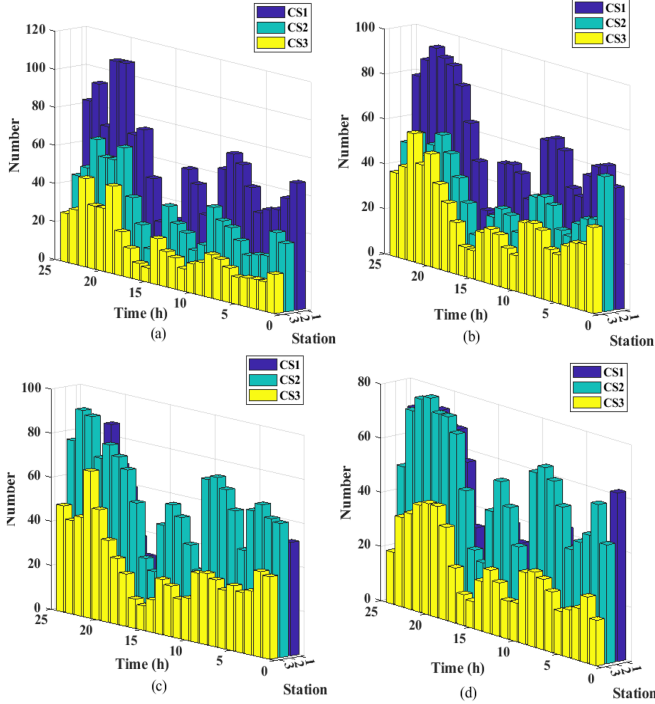


Fig. 4. Number of EV at charging station in case A regarding typical scenarios (a) scenario 1. (b) Scenario 2. (c) Scenario 3. (d) Scenario 4.

TABLE I
THE ASSET INSTALLATION CONDITION OF CASE A

Assets	Installation Conditions
Constructed Branch	19-20, 18-19, 9-17, 8-21, S4-21, 9-22, 22-23, 24-25, 8-25, 8-27, 26-27, 6-28, S4-30, 29-30, 30-43, 37-43, 10-31, 12-45, 44-45, 38-44, 38-39, 32-39, 33-39, 34-35, 35-36, S3-36, S3-41, 40-41, 41-42, 42-48, 48-49, 49-50, 46-47, 14-46
Substation (MVA)	51 (16.7), 52 (13.3), 53 (16.7), 54 (22.2)
PV (MW)	5 (5), 10 (5), 26 (0.9), 30 (2.1), 33 (4.4)
SOP (MVA)	31-37 (0.4), 33-34 (0.5), 2-8 (0.5), 10-15 (0.5)
ESS (MW/MW-h)	8 (1.8, 4.28), 9 (0.9, 1.79), 13 (1.6, 5.17), 16 (0.7, 1.62)
CS (MW)	17 (2), 32 (1.2), 50 (0.8)

by Fig. 3. Additionally, the starting points for EV navigation are set to be different in four scenarios. Hereafter, considering the impact of traffic flow and based on the model of Section II ((1)–(12)), the number of EV and related charging demand at each candidate CS node can be obtained to formulate subsequent planning decisions. With respect to the four typical scenarios, the number of EV at different charging stations for case A is illustrated by Fig. 4. It can be seen that most of the EVs are navigated to charging station 1 and station 2, located at node 8 and 25 of the transportation network, to achieve lowest navigation cost. Moreover, the number of charging EV is peaked at 19–20 o'clock for the three constructed charging stations.

Based on the estimation of EV charging demand, the asset installations of case A can be obtained. And the installation conditions are illustrated in Table I and Fig. 5. To clarify, 51 (16.7) in Table I represents that an asset of 16.7 MW is installed at node 51. Subsequently, the planning decision obtains the

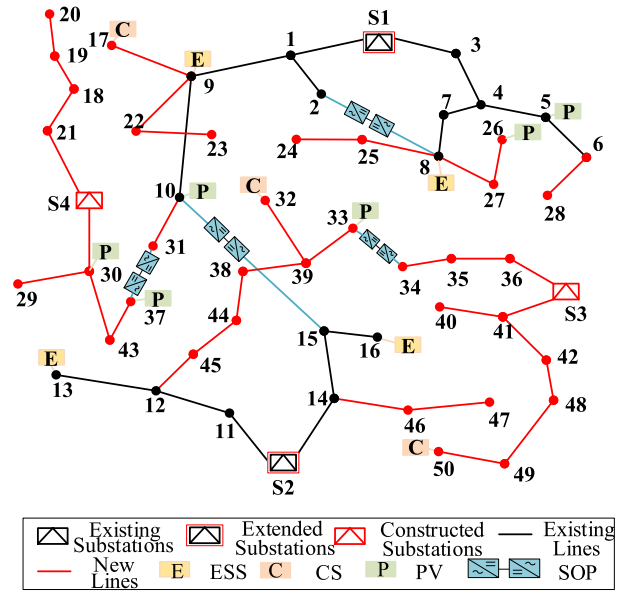


Fig. 5. Planning results of case A.

TABLE II
THE INSTALLATION CONDITION OF CASE B

Assets	Installation Conditions
Constructed Branch	19-20, 18-19, 9-17, 18-21, S4-21, 22-23, 23-24, 24-25, 8-25, 26-27, 27-28, S4-30, 29-30, 37-43, 31-37, 13-43, 12-45, 44-45, 38-39, 32-39, 33-34, 33-39, 34-35, 35-36, S3-36, S3-28, S3-41, 14-46, 16-40, 41-42, 42-48, 48-49, 49-50, 42-47
Substation (MVA)	51 (16.7), 52 (16.7), 53 (22.2), 54 (22.2)
PV (MW)	5 (0.3), 8 (2.3), 10 (1.7), 26 (2.8), 30 (1.5), 33 (2.2), 37 (3)
SOP (MVA)	-
ESS (MW/MW-h)	8 (1.4, 2.71), 9 (0.9, 2.44), 13 (1.6, 5.65), 16 (0.1, 2.03)
CS (MW)	17 (1.8), 32 (1.4), 50 (0.8)

connectivity of the test system. As for deployed SOPs, it can be seen that most of the SOPs are installed between different circuits to provide active/reactive power exchange capability. The installed PV and ESS are mostly located in the nearby area to eliminate the PV curtailment.

B. Expansion Planning Refers to EV Navigation

In case B, the planning approach only considers EV navigation but ignores the deployment of SOP. The asset installation conditions of case B are shown in Table II and Fig. 6. The planning decision also maintains the connectivity. And the lack of SOP deployment changes the installation of other assets. And combined with Table I, it can be seen that the installed PV capacities are more disjointed than in case A.

Moreover, according to the model of Section II, the number of charging EV at three CS stations for case B is different from case A. In this case, more of the EV charging load are deployed to charging station 2, located at node 25, while more than 1/2 of EV load is deployed to charging station 2 and 3. And comparing with case A, it can be concluded that the charging load of EV

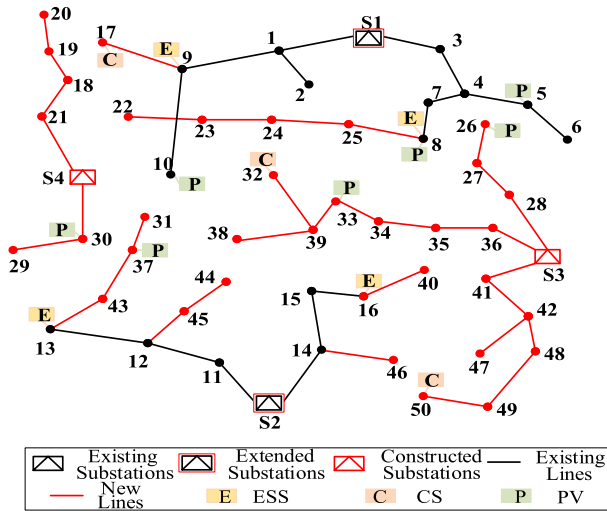


Fig. 6. Planning results of case B.

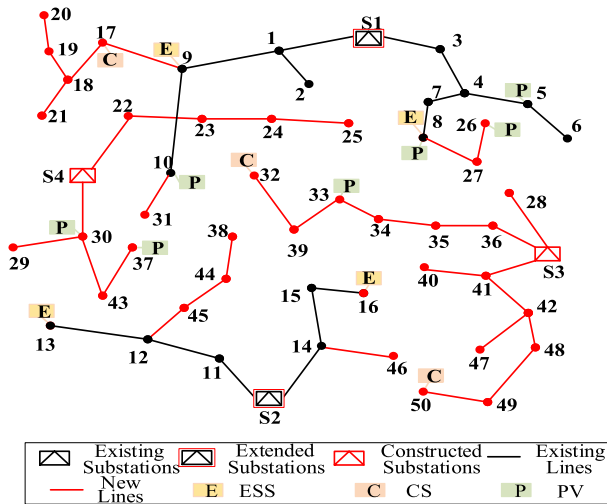


Fig. 7. Planning results of case C.

at peak hours is less coordinated in case B owing to the lack of deployed SOP.

C. Expansion Planning Neglecting EV Navigation and SOP

In case C, a conventional expansion planning without consideration of EV navigation and SOP deployment is proposed. The asset installation conditions of case C are shown in Fig. 7 and Table III. Comparing to case A and case B, the different unset branch construction leads to a variant system structure with more deployed capacities.

And owing to that different structure, although the overall installation capacity of PV, ESS and CS in case C is the same as case B's, the installation at each node obtains different capacities. It is also worth noticed that, comparing to case B, the reason for the higher probability of DSM events in this case would be that the constant EV charging demand leads to lower possibilities to manage the overall load.

TABLE III
THE INSTALLATION CONDITION OF CASE C

Assets	Installation Conditions
Constructed Branch	19-20, 18-19, 17-18, 9-17, 18-21, S4-22, 22-23, 23-24, 24-25, 8-27, 26-27, S4-30, 29-30, 30-43, 37-43, 10-31, 12-45, 44-45, 38-44, 32-39, 33-39, 33-34, 34-35, 35-36, S3-50, S3-28, S3-41, 40-41, 41-42, 42-48, 48-49, 49-50, 42-47, 14-46
Substation (MVA)	51 (16.7), 52 (16.7), 53 (22.2), 54 (22.2)
PV (MW)	5 (0.4), 8 (2.4), 10 (1.7), 26 (2.7), 30 (2), 33 (2.1), 37 (2.6)
SOP (MVA)	-
ESS (MW/MW·h)	8 (1.8, 4.07), 9 (1.6, 3.97), 13 (0.8, 2.49), 16 (0.7, 1.84)
CS (MW)	17 (1.5), 32 (1.3), 50 (1.2)

TABLE IV
THE PLANNING COST OF CASE A, B AND C

Annual Cost (M\$/year)	Case A	Case B	Case C
Total Cost	35.22	38.89	36.75
Branch Investment Cost	0.36	0.32	0.38
Substation Investment Cost	4.53	5.41	5.41
PVG Investment Cost	1.27	1.01	1.01
SOP Investment Cost	0.05	0	0
ESS Investment Cost	0.08	0.08	0.08
CS Investment Cost	0.05	0.05	0.05
Operation Cost	26.67	26.11	26.14
PV Curtailment Cost	0	0.13	0.2
DSM Cost	0	3.42	3.48
EV Navigation Cost	2.21	2.36	0

D. Annual Cost Comparison

The planning costs that include asset investment costs and operation costs are presented in Table IV. In case A, the planning scheme is established based on consideration of SOP and EV navigation. It can be seen that the PV curtailment obtains a zero value which indicates a 100% accommodation of PV generation. Furthermore, owing to the extra flexibility provided by SOP, DSM is not triggered in this case and the DSM cost is 0 then. As for case B, the installation of SOP is not considered. The total cost of planning is still raised by 10 percent comparing to case A. Representing lower renewable energy accommodation, the PV investment is declined while the PV curtailment is raised above \$0.13 M. The triggered DSM relates to a cost of \$3.42 M. Additionally, comparing to case A, the EV navigation cost is raised by 6%. Since case C neglects both SOP deployment and EV navigation, the generated scenarios of navigate EV would be directly seen as the demand of all the charging stations. Subsequently, comparing to case A, the cost except EV navigation of case C is raised by 11%, and the total cost is raised by 4%. The investment cost of substation, ESS and CS is the same as case B while the branch investment obtains different value. And although the PV investment is the same as case B, a comparatively high PV curtailment cost appears in case C. Moreover, the triggered DSM relates to higher cost than case B's. A possible reason for the facts above could be without consideration of EV navigation, more PV power should be curtailed.

E. Summary of the Results

As a result, it can be seen that case A has the lowest cost with the least PV/load curtailment, while case C has the highest. Therefore, it can be concluded that considering EV navigation and the use of SOP in the expansion planning would benefit the mitigation of various costs. Furthermore, the accommodation of renewable energy can also be improved by considering EV navigation and SOP deployment.

V. CONCLUSION

This article proposes an expansion planning approach for SOP based distribution systems considering EV navigation. The model of EV navigation is employed with considerations of traffic flow. Thus, the planning cost and EV navigation cost can be mitigated simultaneously. The subsequent case study analyses the rationality of considering SOP and EV navigation to minimize costs related to expansion planning and EV navigation. On the premises of the analysis above, the key findings are:

- Although SOP installation requires extra budget, the planning cost of distribution systems and EV navigation cost on transportation network can be considerably reduced owing to the flexibility offered by SOP.
- The mitigated PV and load curtailment indicates that the renewable energy accommodation can be enhanced by considering SOP employment and EV navigation.

Subsequently, regarding the distribution system coupled with transportation network, the future work of this research could be investigating the joint planning approach for both the distribution system and the transportation network in terms of traffic flow.

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